



AASMT

Faculty of Engineering

Operating Systems

Lecturer

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Chapter 3: Processes

Chapter 3: Processes

- Process Concept
- Process Control Block (PCB)
- Process Scheduling
- Operations on Processes

Process Concept (Cont.)

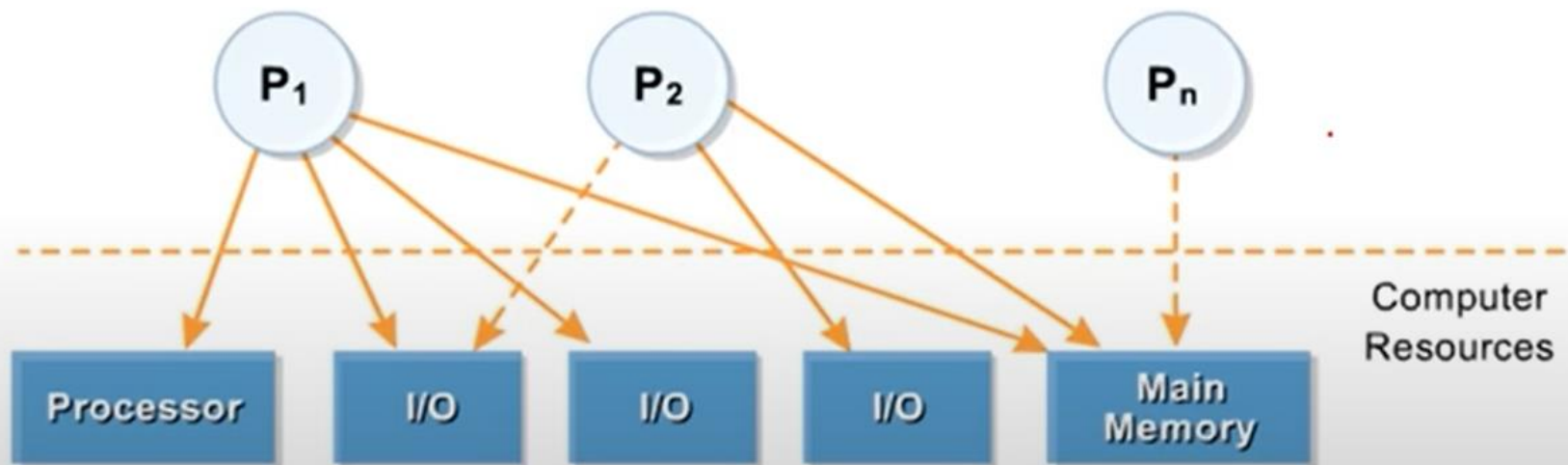
- Program vs. Process
- Program is **passive**, entity such as the file that contains the list of instructions on a disk always referred to as an (**executable file**),
- process is **active**, Program becomes process when executable file loaded into the memory and then becomes an active entity

Process Concept (Cont.)

- The fundamental task of any operating system is the [process management](#)
- Processes include not only a text but also include a set of resources such as open files and pending signals.
- Processes also contain internal kernel data, processor state, an address space, and a data section.

Process Concept (Cont.)

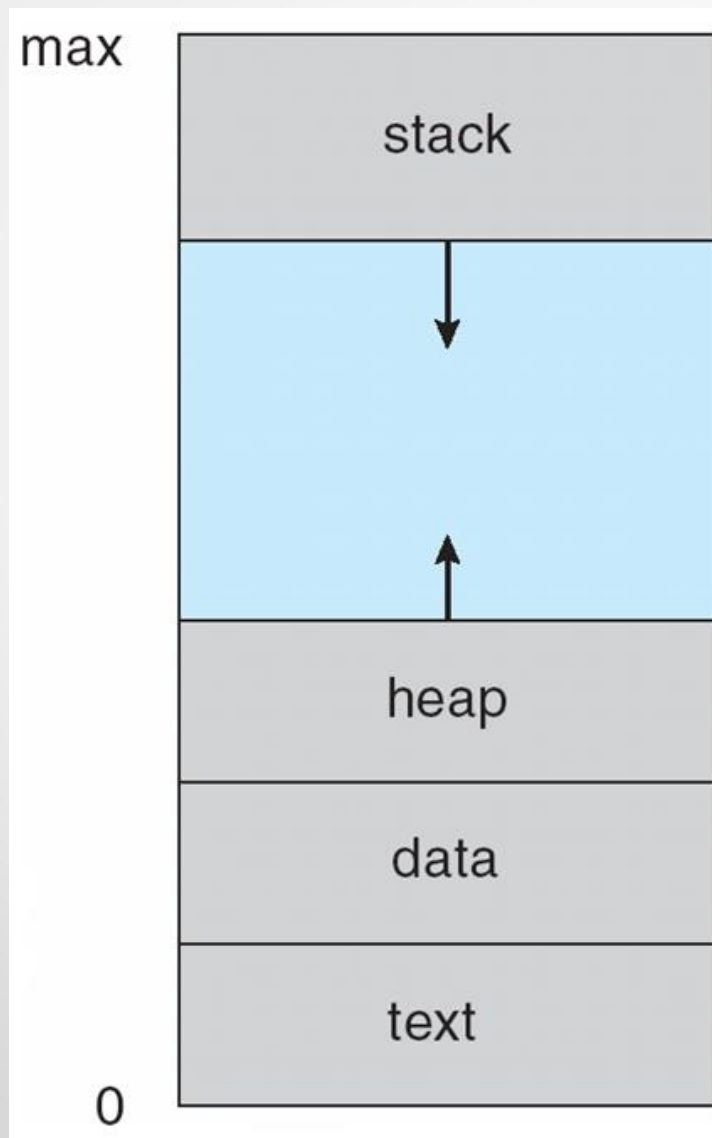
- OS must **allocate resources** to processes, enable sharing of information, protect resources, and enable the synchronization among processes.



Process Elements

- Segments of a process represents the following components:
 - **Text Section:** the program code. This is typically read-only, and might be shared by a number of Processes.
 - **Data Section:** containing global variables.
 - **Heap:** containing memory dynamically allocated during run time.
 - **Stack:** containing temporary data like “function parameters, return address, local variables,”
function parameters, return addresses, local variables

Process in Memory



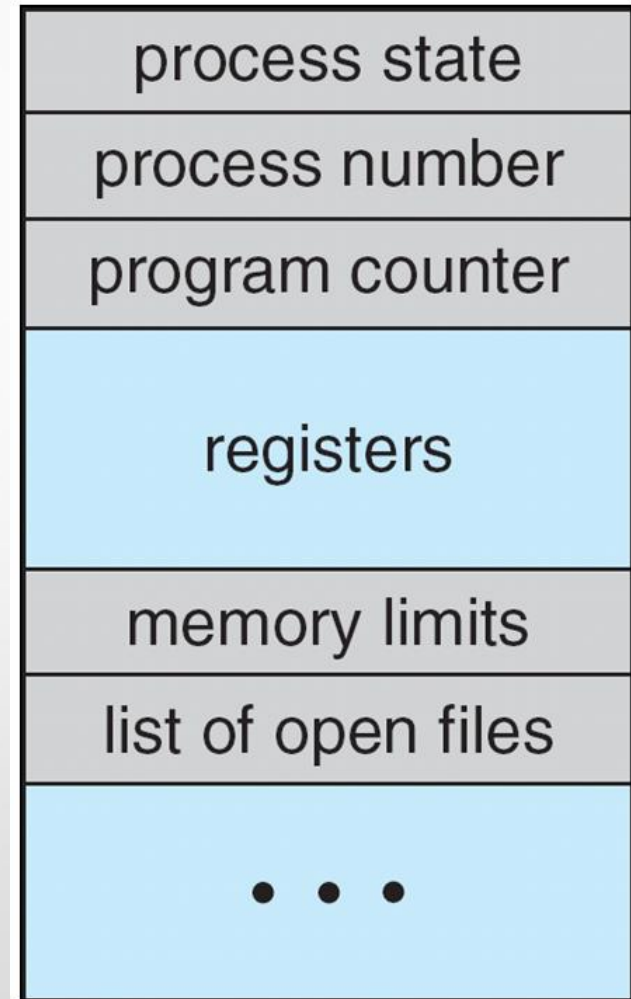
Process Control Block (PCB)

Information associated with each process

(also called Task Control Block)

Process identification information

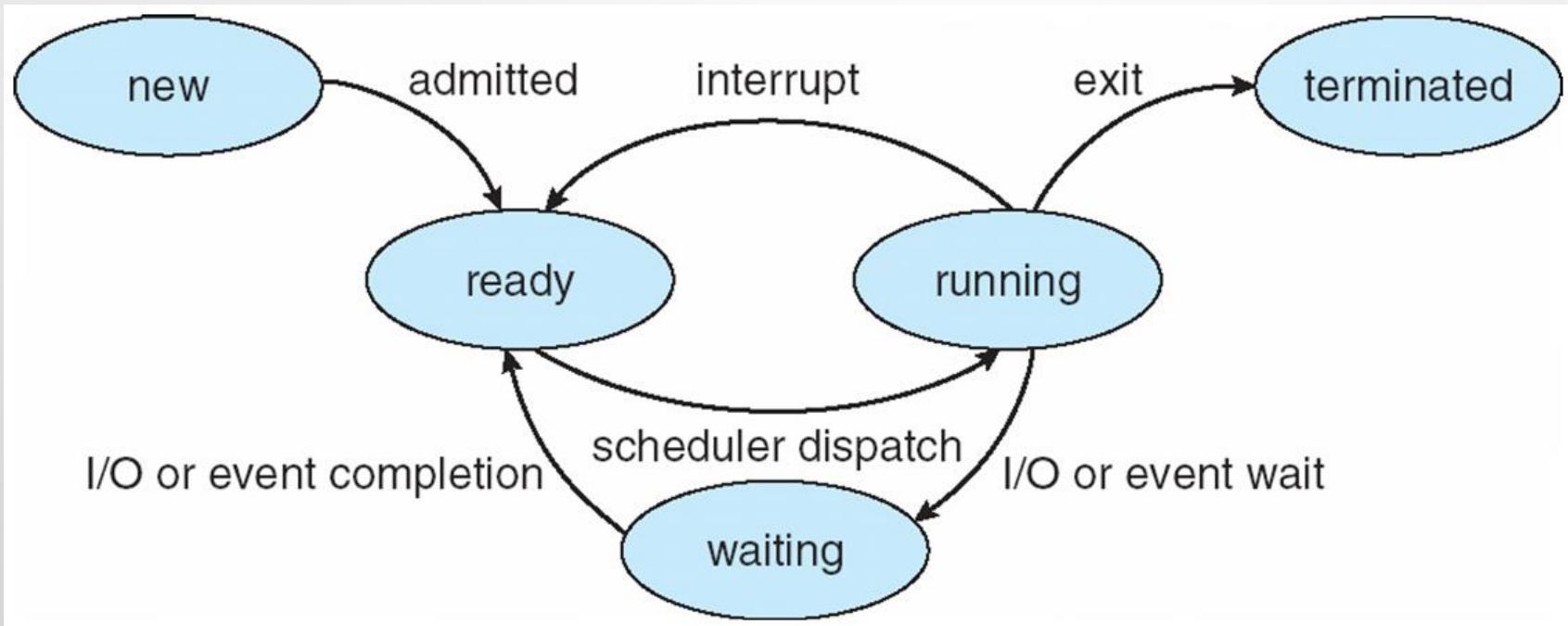
- Process state Information: Processor state, running, waiting, etc
- Program counter: location of instruction to next execute
- CPU registers: contents of all process-centric registers
- CPU scheduling information: priorities, scheduling queue pointers
- Memory-management information: memory allocated to the process
- Accounting information: CPU used, clock time elapsed since start, time limits
- I/O status information: I/O devices allocated to process, list of open files



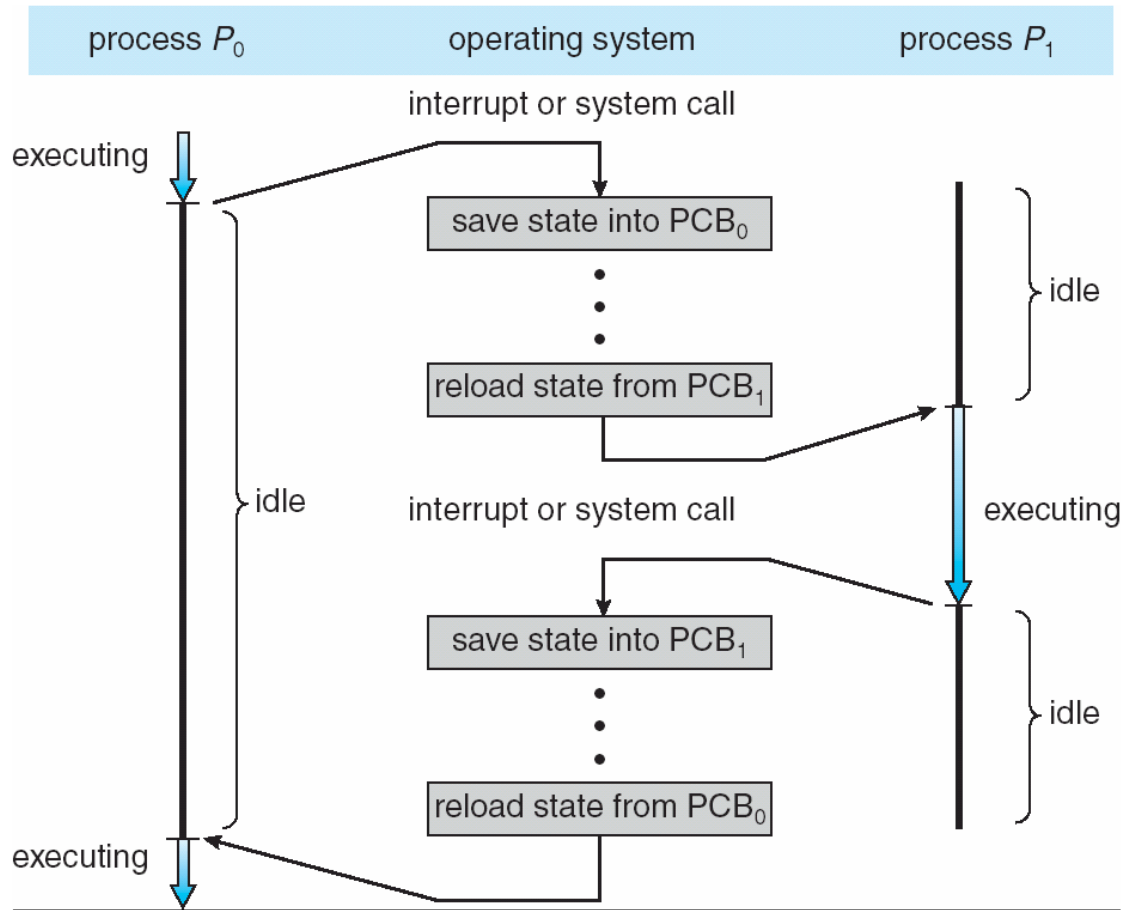
Process State

- As a process executes, it changes **state**
 - **new**: The process is being created
 - **running**: Instructions are being executed
 - **waiting**: The process is waiting for some event to occur
 - **ready**: The process is waiting to be assigned to a processor
 - **terminated**: The process has finished execution

Diagram of Process State

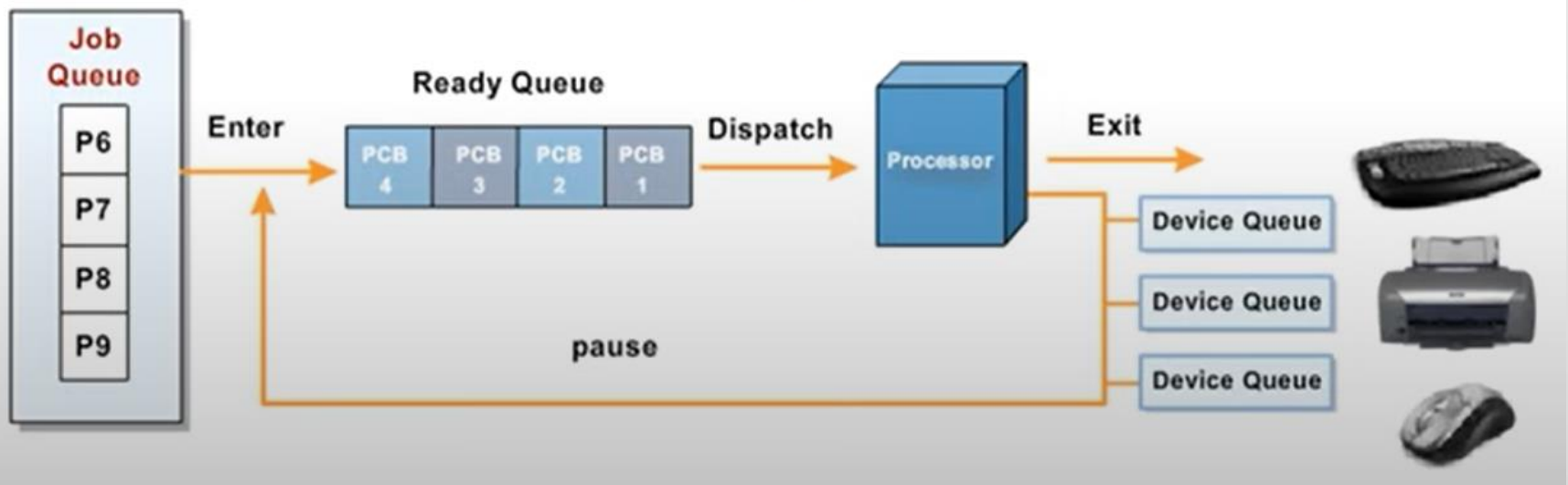


CPU Switch From Process to Process



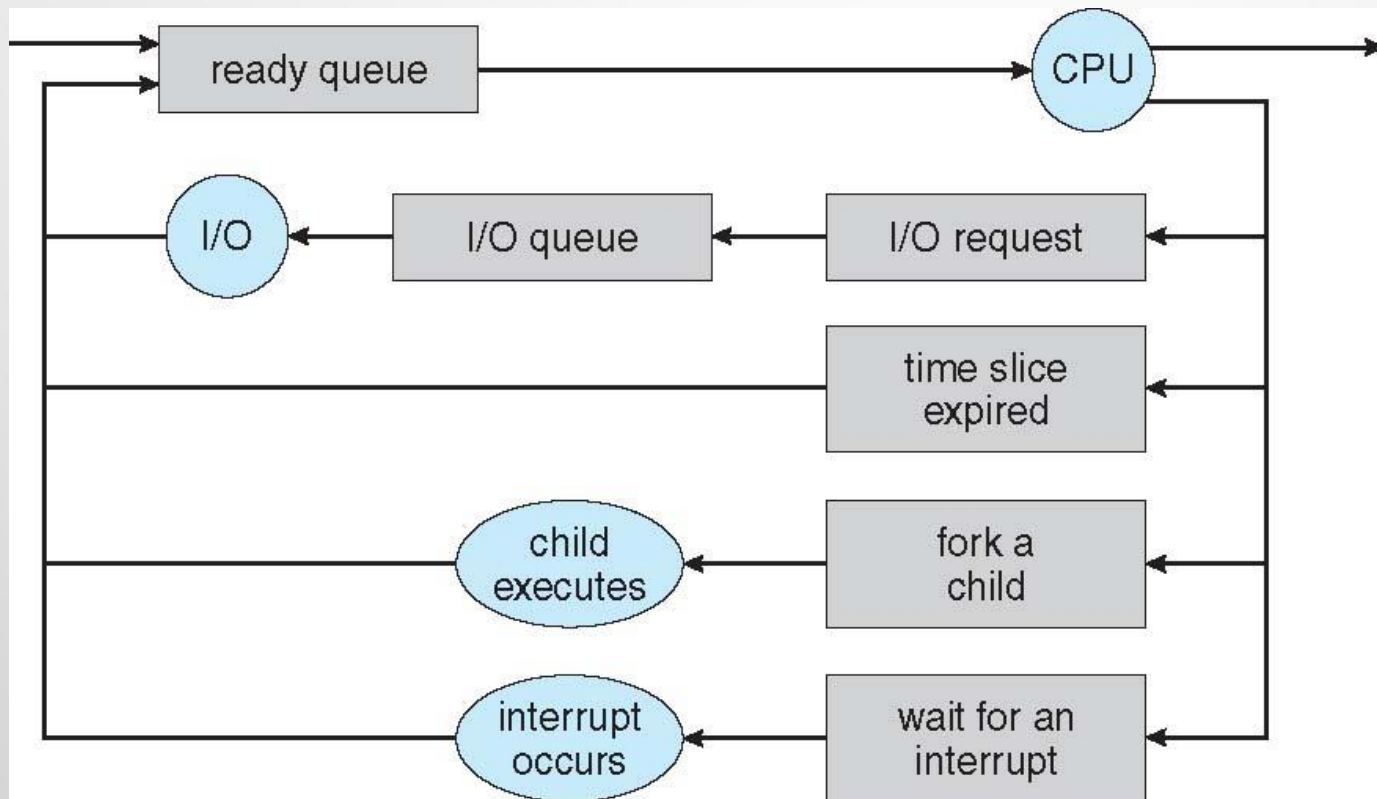
Process Scheduling

- Process scheduler selects among available processes for next execution on CPU
 - Maintains scheduling queues of processes
 - Job queue: set of all processes in the system
 - Ready queue: set of all processes residing in main memory, ready and waiting to execute
 - Device queues – set of processes waiting for an I/O device
- Processes migrate among the various queues



Representation of Process Scheduling

- Queueing diagram represents queues, resources, flows

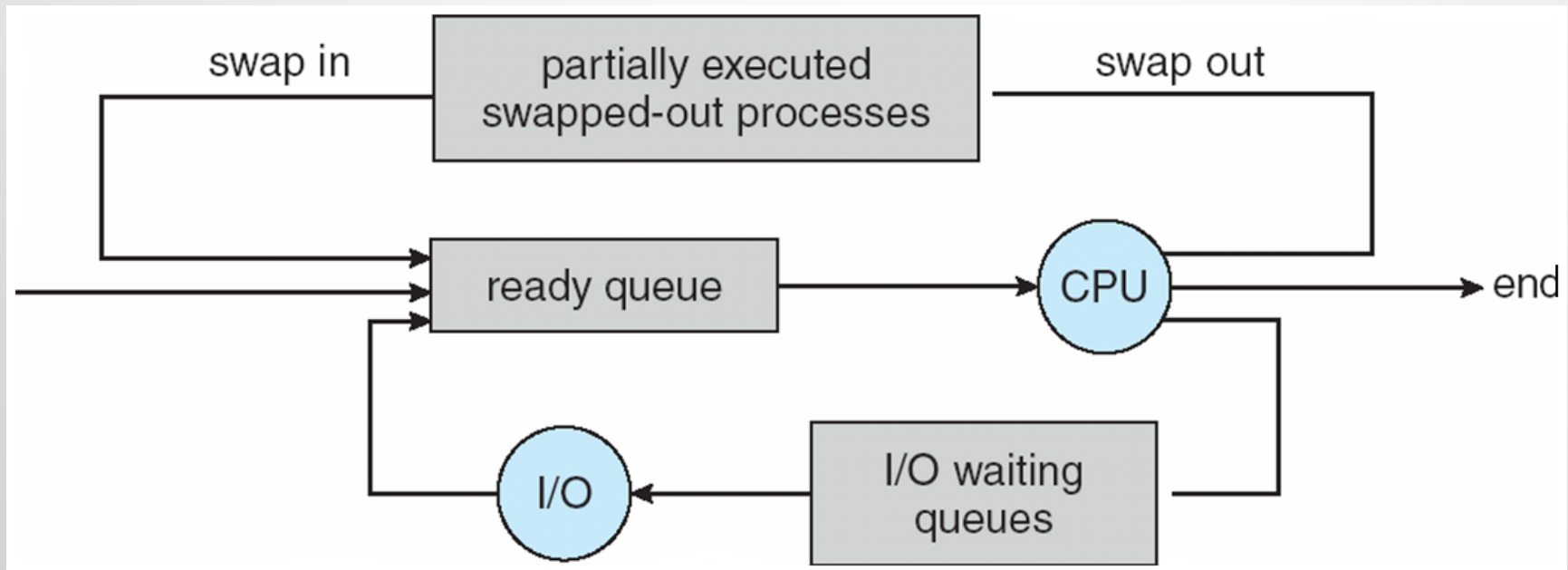


Schedulers

- Short-term scheduler (or CPU scheduler)
 - selects which process should be executed next and allocates CPU
 - Short-term scheduler is invoked frequently (milliseconds) ⇒ (must be fast)
- Long-term scheduler (or job scheduler)
 - selects which processes should be brought into the ready queue
 - Long-term scheduler is invoked infrequently (seconds, minutes) ⇒ (may be slow)
 - The long-term scheduler controls the degree of multiprogramming

Addition of Medium-Term Scheduling

- Medium-term scheduler can be added if degree of multiple programming needs to decrease
 - Remove process from memory, store on disk, bring back in from disk to continue execution: [swapping](#)





End of Chapter 3